Ripple Perception in Musicians Compared to Non-Musicians

Effects of Bandwidth and Intensity

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Abstract

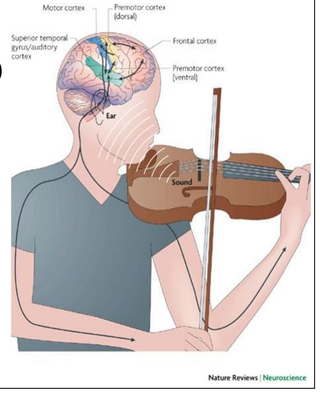
- Noisy environments can be problematic for individuals with hearing loss. Hearing aids amplify sounds, and are often the primary rehabilitation tool. The auditory system, however, is not designed to optimally process speech at high intensities.
- Lateral inhibition (LI) is a hearing mechanism that is designed to enhance speech and music perception at high intensities for individuals with normal hearing.
- This study measured thresholds in musicians and non-musicians using a rippled signal that was parametrically varied to induce lateral inhibition. We hypothesized that lateral inhibition will enhance auditory perception as signal bandwidth and intensity increases.
- Results showed that ripple perception decreased at high intensities. Thresholds were worst for the narrow bandwidth signal and improved with increase in signal bandwidth.
- Musicians performed better for the wider bandwidth signal. Our findings show that musicians present a good model for studying auditory mechanisms related to lateral inhibition. Future work will explore lateral inhibition in individuals with hearing loss.

Background

- Understanding speech in noisy environments can be challenging for everyone. Individuals with normal hearing perform better than those with hearing loss when compared at high intensities.
- Lateral inhibition occurs when exciting a specific neuron results in simultaneous suppression of neighboring neurons, to enhance spectral features at high intensities for individuals with normal hearing.
- This study measured spectral ripple perception thresholds in musicians and non-musicians using a spectral signal that was parametrically varied to induce lateral inhibition. We hypothesized that musicians' performance would show functional benefit of bandwidth at high intensity compared to non-musicians.

Objectives

- We tested bandwidth and intensity effects in musicians and non-musicians to evaluate the following questions:
  1. Do Spectral Modulation Detection(SMD) thresholds degrade at high intensities?
  2. Does bandwidth improve SMD thresholds at high intensities?
  3. Are musicians more robust to high-intensity effects compared to non-musicians?



Results

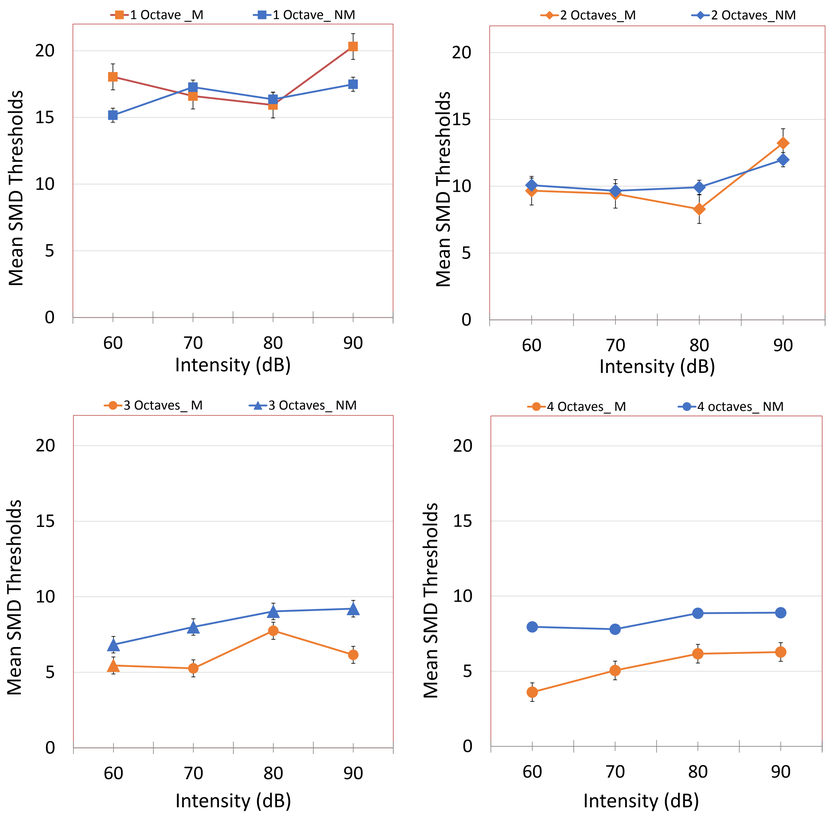


Figure1. Average SMD Thresholds for Musicians and Non-Musicians

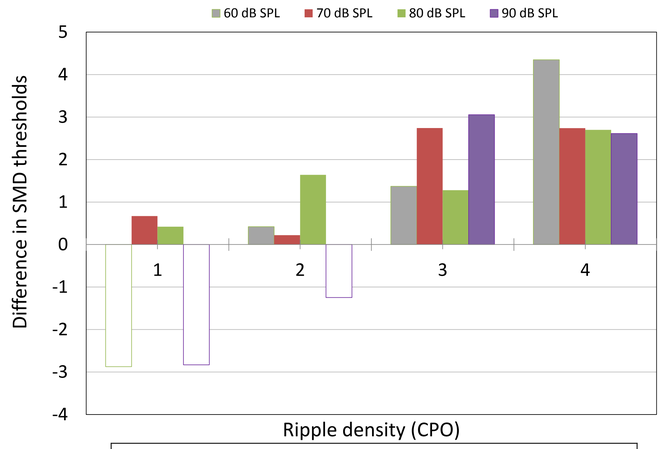


Figure 2. Difference in SMD thresholds in non-musicians and musicians

SUMMARY

1. Ripple thresholds (lower = better) declined with increase in intensity for all bandwidth conditions.

2. At the highest intensity, SMD thresholds were best for the widest bandwidth (i.e. 4 octaves) condition.

3. Musicians had better SMD thresholds for the wider bandwidth conditions (i.e. 3 and 4 octave). They were also more robust to the negative effects of high intensities.

Procedure

- All qualified participants were administered a 3 alternative forced-choice test and asked to identify the interval that sounded different.
- The signal as a Gaussian noise with sinusoidal modulations of 2 cycles per octave (CPO).
- Conditions were ran at four different octave:1,2,3 and 4 octaves.
- The spectral modulation depth (dB) of the test signal was varied adaptively using a two-up, one-down tracking procedure.
- Spectral modulation depth detection threshold (SMD) represented the smallest modulation depth that each listener could detect the presence of the spectral modulated signal.
- Average thresholds were based on three runs for each test condition. Results were extracted from MATLAB and plotted using Microsoft excel.

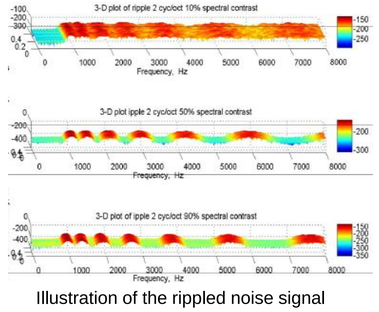


Illustration of the rippled noise signal

Conclusion

- Our results maintained that musicians could discriminate at high intensities better than non-musicians for the wider bandwidth signal compared to the narrow bandwidth signal.
- Our findings confirmed that musicians had behavioral thresholds that were suggestive of greater inhibition compared to non-musicians.
- Future work will explore lateral inhibition in individuals with hearing loss.
- Further exploration of high intensities on musicians and the effects of signal processing on lateral inhibition could help compensate for some of the current limitations of hearing aids.

Acknowledgments

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